

# Majed Alfaraj

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## EDUCATION

### University of California, Berkeley

Berkeley, CA

*B.Sc. Electrical Engineering and Computer Sciences; GPA: 4.0*

*May 2027*

- **Coursework:** Machine Learning, Compilers, Computer Security, Computer Architecture, Data Structures, OS

## EXPERIENCE

### Google

Seattle, WA

*Software Engineering Intern*

*May 2026 – August 2026*

- **Architected** a decoupled multi-agent validation framework in **Go** for Google Cloud Terraform tooling, with stateless orchestration, dynamic judge discovery, and **10+ concurrent workers**.
- **Established** the team-wide validation workflow for new judge LLMs, building **2 evaluation pipelines** and a pluggable judge registry with **Go, Protocol Buffers, and Gemini language server injection** to test routing, failure modes, and hallucinations.
- **Productionised 6,000+ LOC** of reusable evaluation infrastructure, processing **hundreds of code diffs in minutes**, surfacing previously missed defects, and enabling consistent development and qualification of future judge agents.

### University of California, Berkeley

Berkeley, CA

*Teaching Assistant (UCS2), DATA C100*

*August 2025 – Present*

- **Support a 1,200+ student** upper-division data science course through tutoring, office hours, grading, exam administration, and online instruction.
- **Author parameterised exam questions in PrairieLearn** for **1,000+ students**, writing randomised question generators and grading logic that produce individualised exam variants at scale.

## RESEARCH

### University of Toronto | Baycrest Health Sciences

Toronto, Canada

*Co-op Researcher | Cranium Lab; PI: Professor J Jean Chen*

*June 2025 – August 2025*

- **Developed** a 3D conditional GAN in **PyTorch/MONAI** to translate resting-state CBF to CVR from paired MRI volumes, including 3D preprocessing and evaluation pipelines.
- **Achieved**  $r = 0.83$ ,  $R^2 = 0.66$ , and PSNR=24.05 dB; reduced CVR error from **8.6 to 5.4** and doubled SSIM from **0.18 to 0.38**.

## PROJECTS

### Lisp Compiler & Interpreter | OCaml, C, x86-64 Assembly

Aug. 2025 – Dec. 2025

- **Built** a Lisp interpreter and compiler with tagged runtime values, lexical scoping, file I/O, computed jumps, function inlining, and common subexpression elimination.

### PintOS Operating System | C, Rust

Jan. 2026 – May 2026

- **Extended** the PintOS kernel with user processes and system calls, strict-priority scheduling with nested priority donation, user threads, and a concurrent file system with a write-back buffer cache, extensible files, and hierarchical directories.

### Cryptographically Secure File Sharing | Go

Jan. 2025 – May 2025

- **Implemented** an end-to-end encrypted file-sharing system with authenticated access control, secure sharing and revocation, and tamper-resistant storage.

## HONOURS

**Two-Time Bronze Medalist, Saudi National Olympiad in Informatics** | Top 5 nationally

2020, 2021

## TECHNICAL SKILLS

**Languages:** Go, Python, C++, Java, OCaml, SQL, x86-64

**AI/ML & Systems:** Agents, PyTorch, MONAI, Protocol Buffers, GitHub API, Git